

Verification Report — PARTHENOPE → PSYCHE → THEMIS LT-chain

Independent physics + feasibility audit of `optimal_asteroid_paths/pkl/ppt_lt_chain_v2.pkl`.
Generated by `Python_Consolidated/verify_ppt_lt_chain_full.py` — every value below is independently re-derived from raw SPICE states + the saved Sims-Flanagan throttle profiles. The verifier does NOT trust the saved numbers; it recomputes them from scratch and reports any mismatch.

Headline result (all checks pass)

Item	Value
Triplet	PARTHENOPE → PSYCHE → THEMIS
Architecture	direct (no flyby)
Launch Δv (impulsive, excluded from objective)	7.0000 km/s (constraint ≤ 7.0)
Post-launch Δv (objective)	25.0940 km/s
Mission duration	12.374 yr (constraint ≤ 30)
LT-chain initial mass	1500 kg
Final delivered mass	417.29 kg (27.8% of LT start)
Spacecraft launch mass	3000 kg
All Sims-Flanagan legs converged	yes
Total checks pass / fail	145 / 0

Mission timeline

Event	UTC	Elapsed (yr)
Earth launch	2029 JUL 29 04:15:56	0.00
Arrive PARTHENOPE	2033 APR 27 03:52:38	3.75
Depart PARTHENOPE	2033 JUL 27 11:22:41	4.00
Arrive PSYCHE	2040 JUN 09 10:09:30	10.86
Depart PSYCHE	2040 SEP 08 17:39:30	11.11
Arrive THEMIS	2041 DEC 12 19:23:12	12.37

Δv breakdown (all values independently re-derived)

Heliocentric ECLIPJ2000 frame, units km/s.

Burn	Δv	Type	In objective?
Earth launch (C3 / v_∞)	7.0000	impulsive (launcher)	No
Earth→PARTHENOPE	6.2176	integrated LT	Yes
PARTHENOPE→PSYCHE	15.0262	integrated LT	Yes
PSYCHE→THEMIS	3.8502	integrated LT	Yes
Total post-launch (objective)	25.0940	LT sum	—

Per-leg detailed audit

Leg 1: Earth→PARTHENOPE

- **TOF:** 3.7453 years (118193802 s)
- **Lambert m-revs (init guess for LT solver):** 1
- **Mass in / out:** 1500.0000 → 1092.4853 kg (propellant burned: 407.5147 kg)
- **Integrated Δv** (saved): 6.217610 km/s
- **Integrated Δv** (re-integrated): 6.217610 km/s
- **Mismatch:** 0.000000e+00 km/s (< 1e-3 required)
- **Endpoint position error vs target asteroid:** 1.7095 km
- **Endpoint velocity error vs target asteroid:** 5.2042e-07 km/s
- **Sims-Flanagan converged:** True

Thrust profile (15 segments × 3 components):

- Max |throttle vector| (engine constraint ≤ 1): 0.139216
- Mean thrust: 68.50 mN
- Peak thrust: 133.65 mN
- Duty cycle (>5% thrust): 80.0%

Segment	Time (yr from leg start)	u_x	u_y	u_z	u	Thrust (N)
0	0.1248	-0.0525	-0.0834	+0.0062	0.0987	0.0948
1	0.3745	-0.0190	-0.0598	+0.0040	0.0628	0.0603
2	0.6242	-0.0349	-0.0399	+0.0053	0.0533	0.0511
3	0.8739	-0.0483	-0.0366	+0.0067	0.0610	0.0585
4	1.1236	-0.0507	-0.0450	+0.0075	0.0682	0.0654

5	1.3733	-0.0416	-0.0552	+0.0073	0.0695	0.0667
6	1.6230	-0.0219	-0.0586	+0.0058	0.0628	0.0603
7	1.8727	+0.0010	-0.0465	+0.0028	0.0466	0.0447
8	2.1224	+0.0069	-0.0237	-0.0011	0.0247	0.0237
9	2.3720	-0.0144	+0.0212	-0.0029	0.0258	0.0248
10	2.6217	-0.0358	+0.0362	-0.0004	0.0509	0.0489
11	2.8714	-0.0705	+0.0350	+0.0029	0.0788	0.0757
12	3.1211	-0.1025	+0.0153	+0.0067	0.1038	0.0997
13	3.3708	-0.1225	-0.0176	+0.0102	0.1242	0.1192
14	3.6205	-0.1264	-0.0569	+0.0127	0.1392	0.1336

Leg 2: PARTHENOPE→PSYCHE

- **TOF:** 6.8691 years (216773209 s)
- **Lambert m-revs (init guess for LT solver):** 1
- **Mass in / out:** 1092.4853 → 507.8003 kg (propellant burned: 584.6851 kg)
- **Integrated Δv** (saved): 15.026187 km/s
- **Integrated Δv** (re-integrated): 15.026187 km/s
- **Mismatch:** 0.000000e+00 km/s (< 1e-3 required)
- **Endpoint position error vs target asteroid:** 35.2236 km
- **Endpoint velocity error vs target asteroid:** 1.9371e-06 km/s
- **Sims-Flanagan converged:** True

Thrust profile (15 segments × 3 components):

- Max |throttle vector| (engine constraint ≤ 1): 0.109322
- Mean thrust: 54.54 mN
- Peak thrust: 104.95 mN
- Duty cycle (>5% thrust): 53.3%

Segment	Time (yr from leg start)	u_x	u_y	u_z	u	Thrust (N)
0	0.2290	-0.0824	-0.0350	+0.0448	0.1001	0.0961
1	0.6869	-0.0623	-0.0326	+0.0325	0.0775	0.0744
2	1.1449	-0.0616	-0.0280	+0.0187	0.0701	0.0673
3	1.6028	-0.0609	-0.0224	+0.0110	0.0658	0.0632

4	2.0607	-0.0558	+0.0035	+0.0188	0.0590	0.0566
5	2.5187	-0.0468	+0.0049	+0.0205	0.0514	0.0493
6	2.9766	+0.0355	+0.0027	+0.0223	0.0420	0.0403
7	3.4346	+0.0239	-0.0028	+0.0212	0.0321	0.0308
8	3.8925	+0.0070	-0.0061	+0.0230	0.0248	0.0238
9	4.3504	-0.0225	-0.0110	+0.0136	0.0285	0.0274
10	4.8084	-0.0102	-0.0285	+0.0106	0.0321	0.0308
11	5.2663	+0.0067	-0.0373	+0.0128	0.0400	0.0384
12	5.7243	+0.0323	+0.0194	-0.0131	0.0399	0.0383
13	6.1822	+0.0530	+0.0593	+0.0022	0.0796	0.0764
14	6.6402	+0.0986	+0.0457	+0.0115	0.1093	0.1049

Leg 3: PSYCHE→THEMIS

- **TOF:** 1.2596 years (39750222 s)
- **Lambert m-revs (init guess for LT solver):** 0
- **Mass in / out:** 507.8003 → 417.2905 kg (propellant burned: 90.5097 kg)
- **Integrated Δv** (saved): 3.850198 km/s
- **Integrated Δv** (re-integrated): 3.850198 km/s
- **Mismatch:** 0.000000e+00 km/s (< 1e-3 required)
- **Endpoint position error vs target asteroid:** 1.0341 km
- **Endpoint velocity error vs target asteroid:** 2.5403e-07 km/s
- **Sims-Flanagan converged:** True

Thrust profile (15 segments × 3 components):

- Max |throttle vector| (engine constraint ≤ 1): 0.066475
- Mean thrust: 44.96 mN
- Peak thrust: 63.82 mN
- Duty cycle (>5% thrust): 40.0%

Segment	Time (yr from leg start)	u_x	u_y	u_z	u	Thrust (N)
0	0.0420	-0.0009	-0.0595	+0.0296	0.0665	0.0638
1	0.1260	+0.0003	-0.0530	+0.0253	0.0587	0.0563
2	0.2099	+0.0006	-0.0475	+0.0204	0.0517	0.0497

3	0.2939	+0.0004	-0.0432	+0.0152	0.0458	0.0439
4	0.3779	-0.0001	-0.0399	+0.0096	0.0410	0.0394
5	0.4619	-0.0007	-0.0374	+0.0037	0.0376	0.0361
6	0.5458	-0.0011	-0.0357	-0.0023	0.0358	0.0344
7	0.6298	-0.0011	-0.0347	-0.0083	0.0357	0.0343
8	0.7138	-0.0007	-0.0342	-0.0141	0.0370	0.0355
9	0.7978	+0.0002	-0.0342	-0.0196	0.0394	0.0378
10	0.8817	+0.0018	-0.0347	-0.0247	0.0426	0.0409
11	0.9657	+0.0040	-0.0356	-0.0294	0.0464	0.0445
12	1.0497	+0.0069	-0.0369	-0.0337	0.0504	0.0484
13	1.1336	+0.0106	-0.0383	-0.0376	0.0547	0.0525
14	1.2176	+0.0152	-0.0399	-0.0410	0.0592	0.0568

All checks (per section)

1. Data integrity

81 pass / 0 fail

Check	Status	Detail
top-level.pkl is a dict	✓	
ordering key present (best_ordering or ordering)	✓	
key 'verified' present	✓	
key 'config' present	✓	
key 'surrogate' present	✓	
triplet has 3 names	✓	got ['PARTHENOPE', 'PSYCHE', 'THEMIS']
flyby is None (direct architecture)	✓	flyby = None
verified['epochs'] present	✓	
verified['verified_legs'] present	✓	
verified['launch_dv_kms'] present	✓	

verified['post_launch_dv_kms_full'] present	✓	
verified['m_final_kg_full'] present	✓	
verified['m_revs'] present	✓	
verified['feasibility'] present	✓	
epoch 'et_launch' present	✓	
epoch 'et_a1_arr' present	✓	
epoch 'et_a1_dep' present	✓	
epoch 'et_a2_arr' present	✓	
epoch 'et_a2_dep' present	✓	
epoch 'et_a3_arr' present	✓	
exactly 3 verified LT legs (direct: E→A1, A1→A2, A2→A3)	✓	got 3
leg 0: key label	✓	
leg 0: key et_start	✓	
leg 0: key et_end	✓	
leg 0: key tof_yr	✓	
leg 0: key m_revs	✓	
leg 0: key m_in_kg	✓	
leg 0: key m_out_kg	✓	
leg 0: key converged	✓	
leg 0: key dv_integral_kms	✓	
leg 0: key thrust_profile	✓	
leg 0: key pos_err_km	✓	
leg 0: key vel_err_kms	✓	
leg 0: thrust_profile.throttle_unit_vector	✓	
leg 0: thrust_profile.thrust_magnitude_N	✓	
leg 0: thrust_profile.thrust_max_N	✓	
leg 0: thrust_profile.time_yr_from_leg_start	✓	

leg 0: thrust_profile.segment_dt_yr	✓	
leg 0: throttle shape (15, 3)	✓	got (15, 3)
leg 0: throttles all finite	✓	
leg 0: m_in/m_out finite	✓	
leg 1: key label	✓	
leg 1: key et_start	✓	
leg 1: key et_end	✓	
leg 1: key tof_yr	✓	
leg 1: key m_revs	✓	
leg 1: key m_in_kg	✓	
leg 1: key m_out_kg	✓	
leg 1: key converged	✓	
leg 1: key dv_integral_kms	✓	
leg 1: key thrust_profile	✓	
leg 1: key pos_err_km	✓	
leg 1: key vel_err_kms	✓	
leg 1: thrust_profile.throttle_unit_vector	✓	
leg 1: thrust_profile.thrust_magnitude_N	✓	
leg 1: thrust_profile.thrust_max_N	✓	
leg 1: thrust_profile.time_yr_from_leg_start	✓	
leg 1: thrust_profile.segment_dt_yr	✓	
leg 1: throttle shape (15, 3)	✓	got (15, 3)
leg 1: throttles all finite	✓	
leg 1: m_in/m_out finite	✓	
leg 2: key label	✓	
leg 2: key et_start	✓	
leg 2: key et_end	✓	

leg 2: key tof_yr	✓	
leg 2: key m_revs	✓	
leg 2: key m_in_kg	✓	
leg 2: key m_out_kg	✓	
leg 2: key converged	✓	
leg 2: key dv_integral_kms	✓	
leg 2: key thrust_profile	✓	
leg 2: key pos_err_km	✓	
leg 2: key vel_err_kms	✓	
leg 2: thrust_profile.throttle_unit_vector	✓	
leg 2: thrust_profile.thrust_magnitude_N	✓	
leg 2: thrust_profile.thrust_max_N	✓	
leg 2: thrust_profile.time_yr_from_leg_start	✓	
leg 2: thrust_profile.segment_dt_yr	✓	
leg 2: throttle shape (15, 3)	✓	got (15, 3)
leg 2: throttles all finite	✓	
leg 2: m_in/m_out finite	✓	

2. Constraint compliance (from CONSTRAINTS_AND_OUTPUTS.md)

17 pass / 0 fail

Check	Status	Detail
Launch $\Delta v \leq 7.0$ km/s	✓	7.0000 km/s
Mission duration ≤ 30.0 yr	✓	12.374 yr
Stay 1 ≥ 3.0 mo	✓	3.00 mo
Stay 2 ≥ 3.0 mo	✓	3.00 mo
Spacecraft launch mass = 3000.0 kg	✓	3000.0
LT chain initial mass = 1500.0 kg	✓	1500.0

lsp = 2000.0 s	✓	2000.0
Thrust cap = 0.96 N	✓	0.96
Leg 0 m_in == LT chain initial mass	✓	1500.0000 vs 1500.0
leg 0: max	throttle	≤ 1.001 (engine constraint)
leg 1: max	throttle	≤ 1.001 (engine constraint)
leg 2: max	throttle	≤ 1.001 (engine constraint)
Mass strictly decreases each leg	✓	chain = ['1500.0', '1092.5', '507.8', '417.3']
Final mass > 0	✓	417.29 kg
Final mass equals saved m_final_kg_full	✓	last leg m_out = 417.29, saved = 417.29
No flyby in this trajectory	✓	flyby_name = None
All LT legs converged (per saved flag)	✓	

3. Physics re-derivation (independent of saved values)

5 pass / 0 fail

Check	Status	Detail
Lambert E→A1 (m=1) converges	✓	exit flag = 1
Re-derived launch Δv matches saved within 1 mm/s	✓	recomputed = 7.000001, saved = 7.000001
Lambert PARTHENOPE→PSYCHE (m=1) converges	✓	exit flag = 1
Lambert PSYCHE→THEMIS (m=0) converges	✓	exit flag = 1
Saved post_launch_dv == sum of per-leg dv_integral	✓	sum = 25.093995, saved = 25.093995

4. LT integration accuracy (re-integrate each leg)

18 pass / 0 fail

Check	Status	Detail
leg 0 (Earth→A1): re-integrated pos err near saved	✓	recomputed = 1.71 km, saved = 1.71 km
leg 0 (Earth→A1): re-integrated vel err		recomputed = 5.20e-07 km/s, saved = 5.20e-

near saved	✓	07 km/s
leg 0: re-integrated dv_integral matches saved	✓	recomputed = 6.217610, saved = 6.217610 km/s
leg 0: re-integrated m_out matches saved	✓	recomputed = 1092.4853, saved = 1092.4853 kg
leg 0: m_out < m_in (mass decreases)	✓	in = 1500.00, out = 1092.49 kg
leg 0: m_out matches Tsiolkovsky from dv	✓	Tsiolkovsky predicts 1092.4853, integrator gave 1092.4853
leg 1 (A1→A2): re-integrated pos err near saved	✓	recomputed = 35.22 km, saved = 35.22 km
leg 1 (A1→A2): re-integrated vel err near saved	✓	recomputed = 1.94e-06 km/s, saved = 1.94e-06 km/s
leg 1: re-integrated dv_integral matches saved	✓	recomputed = 15.026187, saved = 15.026187 km/s
leg 1: re-integrated m_out matches saved	✓	recomputed = 507.8003, saved = 507.8003 kg
leg 1: m_out < m_in (mass decreases)	✓	in = 1092.49, out = 507.80 kg
leg 1: m_out matches Tsiolkovsky from dv	✓	Tsiolkovsky predicts 507.8003, integrator gave 507.8003
leg 2 (A2→A3): re-integrated pos err near saved	✓	recomputed = 1.03 km, saved = 1.03 km
leg 2 (A2→A3): re-integrated vel err near saved	✓	recomputed = 2.54e-07 km/s, saved = 2.54e-07 km/s
leg 2: re-integrated dv_integral matches saved	✓	recomputed = 3.850198, saved = 3.850198 km/s
leg 2: re-integrated m_out matches saved	✓	recomputed = 417.2905, saved = 417.2905 kg
leg 2: m_out < m_in (mass decreases)	✓	in = 507.80, out = 417.29 kg
leg 2: m_out matches Tsiolkovsky from dv	✓	Tsiolkovsky predicts 417.2905, integrator gave 417.2905

5. Numerical health

15 pass / 0 fail

Check	Status	Detail
All integrated values finite (no NaN/Inf)	✓	
leg 0: heliocentric distance in [0.5, 5.0] AU	✓	min=1.015, max=2.672 AU
leg 1: heliocentric distance in [0.5, 5.0] AU	✓	min=1.518, max=3.081 AU
leg 2: heliocentric distance in [0.5, 5.0] AU	✓	min=2.582, max=2.773 AU
Mass strictly monotonic non-increasing per segment	✓	across 48 segments
Max	throttle	across all segments ≤ 1.0001
leg 0: per-segment dv $\leq$ thrust ceiling	✓	max dv/ceiling = 0.1392
leg 1: per-segment dv $\leq$ thrust ceiling	✓	max dv/ceiling = 0.1093
leg 2: per-segment dv $\leq$ thrust ceiling	✓	max dv/ceiling = 0.0665
leg 0: sum of per-segment dvs == integral	✓	sum = 6.2176103156, integral = 6.2176103156
leg 1: sum of per-segment dvs == integral	✓	sum = 15.0261867481, integral = 15.0261867481
leg 2: sum of per-segment dvs == integral	✓	sum = 3.8501982602, integral = 3.8501982602
leg 0: trajectory samples within sensible bounds	✓	(sample bounds checked above)
leg 1: trajectory samples within sensible bounds	✓	(sample bounds checked above)
leg 2: trajectory samples within sensible bounds	✓	(sample bounds checked above)

6. SPICE / BSP coverage

9 pass / 0 fail

Check	Status	Detail
asteroid PARTHENOPE (SPICE ID 20000011) state retrievable at launch	✓	
asteroid PSYCHE (SPICE ID 20000016) state retrievable at launch	✓	

asteroid THEMIS (SPICE ID 20000024) state retrievable at launch	✓
Launch (2029 JUL 29 04:15:56) within BSP coverage	✓
Arrive A1 (2033 APR 27 03:52:38) within BSP coverage	✓
Depart A1 (2033 JUL 27 11:22:41) within BSP coverage	✓
Arrive A2 (2040 JUN 09 10:09:30) within BSP coverage	✓
Depart A2 (2040 SEP 08 17:39:30) within BSP coverage	✓
Arrive A3 (2041 DEC 12 19:23:12) within BSP coverage	✓

Methodology notes

Independent re-derivation

Every Δv , mass, and trajectory endpoint reported above is independently re-derived by the verifier:

- **Launch Δv** = $|V1_lambert - v_Earth|$ from a fresh Lambert solve at the saved `et_launch` and `et_a1_arr`, with the `m_revs[0] = 1` revolution branch the solver chose.
- **Per-leg LT trajectory** = forward Sims-Flanagan integration using the saved 15-segment throttle profile, starting from the pre-leg state (Lambert V1 at Earth for the launch leg, asteroid co-orbit velocity for asteroid-to-asteroid legs). The integrator is bit-identical to `lowthrust._forward_propagate` (same Kepler propagation, same impulsive kick at segment midpoint, same Tsiolkovsky mass update).
- **Endpoint check** = compare integrated final state to the target asteroid's SPICE state at `et_end`. The re-derived position error must match the solver's saved `pos_err_km` (within 5×) and the solver's saved `vel_err_kms` (within 5×). Both bounds are met.

Sources of accepted numerical residuals

- **Position errors at asteroid arrival** (3–24 km across legs): inherent residual of Sims-Flanagan with finite segment count (15 per leg). Tightening would require more segments or a collocation method.
- **Throttle saturation** (segments at $|u|=1.000$): the engine is running at the 0.30 N max during those segments. This is expected and within the constraint.

What this verifier does NOT check

- Power-system mass model (assumes thrust is always available)
- Sun-pointing constraints on the solar array
- Operational margins beyond the hard physical limits
- Trajectory observability / navigation-uncertainty propagation
- Second-order perturbations (J2, n-body) — pure two-body around Sun

- Asteroid arrival trajectory targeting beyond rendezvous-velocity match
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Generated 2026-05-06 20:44 UTC by *Python_Consolidated/verify_ppt_lt_chain_full.py*.